An encoder is a combinational logic circuit that converts 2^n input lines into n binary output lines

4-to-2 Encoder:

A **4-to-2 encoder** has **4 inputs** $(D_0, D_1, D_2, D_3)$and **2 outputs** (Y_1, Y_0).

Truth table:

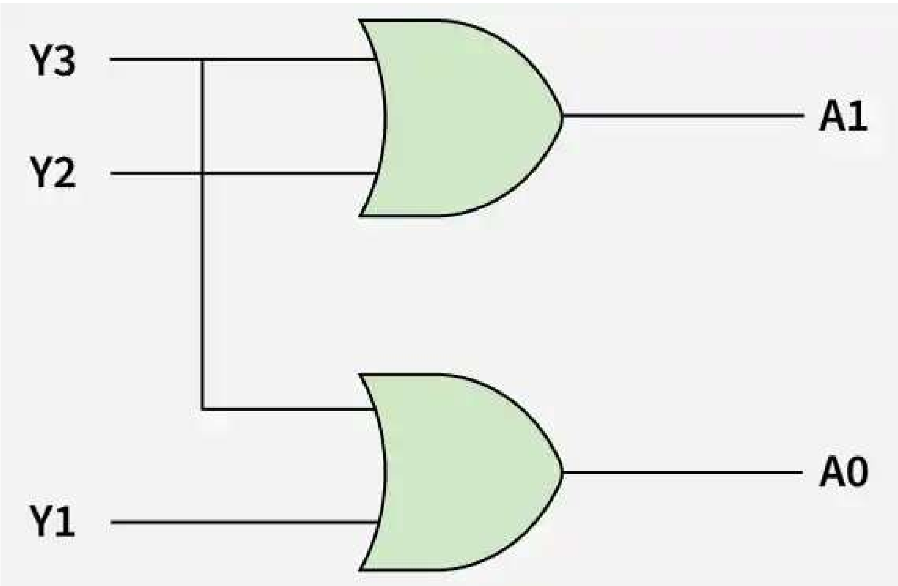
D_0	D_1	D_2	D_3	Y_1	Y_0
1	0	0	0	0	0
0	1	0	0	0	1
0	0	1	0	1	0
0	0	0	1	1	1

Boolean Equations

From the truth table:

$$Y_1 = D_2 + D_3$$

$$Y_0 = D_1 + D_3$$



8 To 3 Encoder

An 8-to-3 encoder has 8 input lines (D_0 to D_7)and 3 output lines (Y_2, Y_1, Y_0).

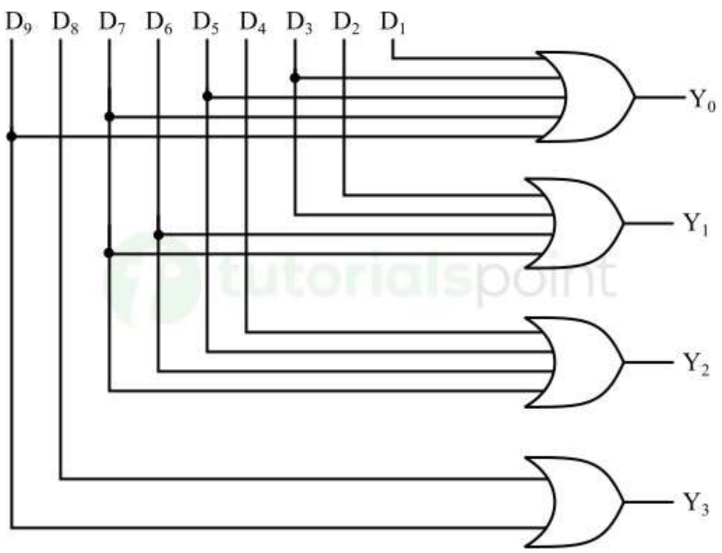
D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	Y_2	Y_1	Y_0
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

Simplified Equation

$$Y_2 = D_4 + D_5 + D_6 + D_7$$

$$Y_1 = D_2 + D_3 + D_6 + D_7$$

$$Y_0 = D_1 + D_3 + D_5 + D_7$$



BCD Encoder

A BCD encoder is a combinational logic circuit that converts decimal digits (0–9) into their corresponding 4-bit BCD (Binary Coded Decimal) code.

BCD Encoder:

$$\boxed{\text{10 inputs} \rightarrow \text{4 outputs}}$$

Decimal Input	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	$Y_3Y_2Y_1Y_0$
0	1	0	0	0	0	0	0	0	0	0	0000
1	0	1	0	0	0	0	0	0	0	0	0001
2	0	0	1	0	0	0	0	0	0	0	0010
3	0	0	0	1	0	0	0	0	0	0	0011
4	0	0	0	0	1	0	0	0	0	0	0100
5	0	0	0	0	0	1	0	0	0	0	0101
6	0	0	0	0	0	1	0	0	0	0	0110
7	0	0	0	0	0	0	0	1	0	0	0111
8	0	0	0	0	0	0	0	0	1	0	1000
9	0	0	0	0	0	0	0	0	0	1	1001

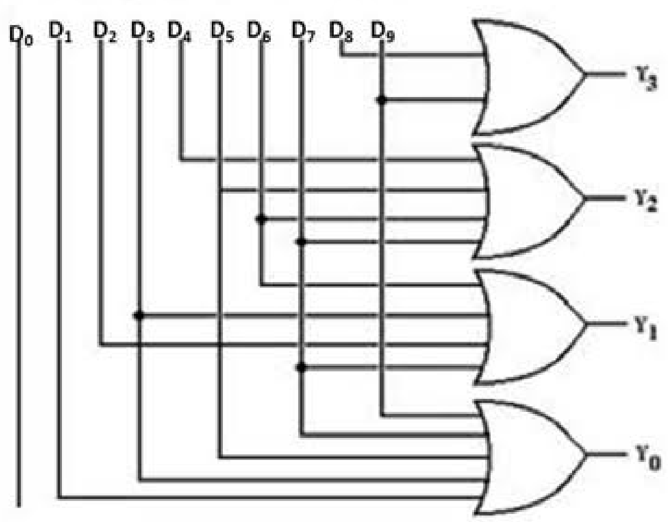
These equations define the Boolean output logic for a **Decimal-to-BCD (Binary-Coded Decimal) Encoder**.

$$Y_4 = D_8 + D_9$$

$$Y_3 = D_4 + D_5 + D_6 + D_7$$

$$Y_2 = D_2 + D_3 + D_6 + D_7$$

$$Y_1 = D_1 + D_3 + D_5 + D_7 + D_9$$



Decoder:

A decoder is a combinational logic circuit that converts $n$input lines into 2^noutput lines, where one output line is activated for each input combination.

2-to-4 Decoder

A 2-to-4 decoder has 2 input lines and 4 output lines.

Let the inputs be A, B and outputs be Y_0, Y_1, Y_2, Y_3.

A	B	Y_0	Y_1	Y_2	Y_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

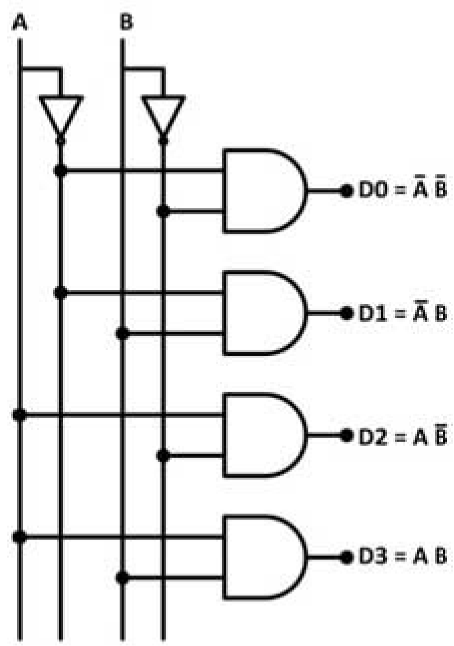
these equations define the Boolean output logic for a 2-to-4 Line Decoder with active-high outputs.

$$Y_0 = \bar{A}\bar{B}$$

$$Y_1 = \bar{A}B$$

$$Y_2 = A\bar{B}$$

$$Y_3 = AB$$



3-to-8 Decoder

A 3-to-8 decoder has 3 input lines and 8 output lines.

$$\boxed{n \text{ inputs} \rightarrow 2^n \text{ outputs}}$$

For $n = 3$:

$$2^3 = 8$$

Let the inputs be A, B, C and outputs be $Y_0$to Y_7.

A	B	C	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

these equations define the Boolean output logic for a 3-to-8 Line Decoder with active-high outputs.

$$Y_0 = \bar{A}\bar{B}\bar{C}$$

$$Y_1 = \bar{A}\bar{B}C$$

$$Y_2 = \bar{A}B\bar{C}$$

$$Y_3 = \bar{A}BC$$

$$Y_4 = A\bar{B}\bar{C}$$

$$Y_5 = A\bar{B}C$$

$$Y_6 = AB\bar{C}$$

$$Y_7 = ABC$$

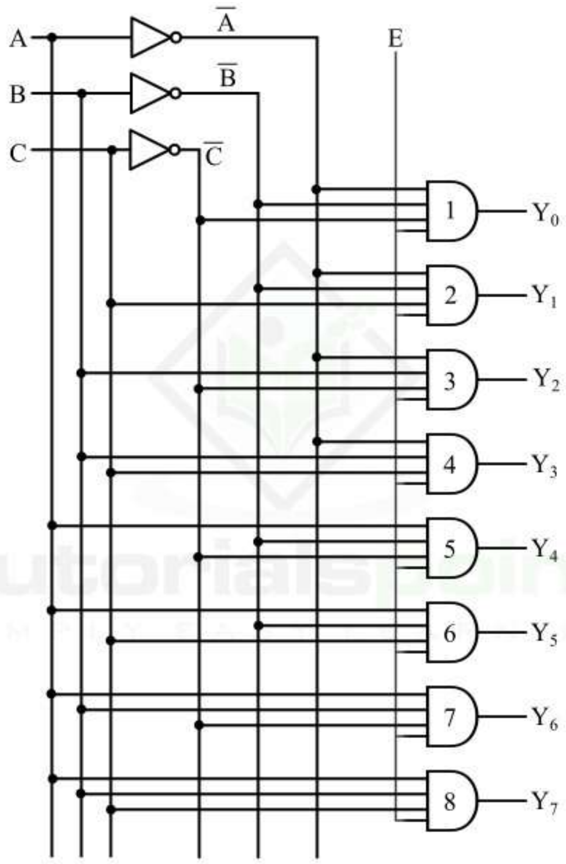


Figure 5 - 3 to 8 Decoder

Multiplexer

multiplexer is a combinational logic circuit that selects one input from multiple input lines and sends the selected input to a single output line**.**

Multiplexer = Many inputs → One output

For a MUX with n selection lines:

$$2^n \text{ input lines} \rightarrow 1 \text{ output}$$

Example: A 4-to-1 MUX has:

- 4 input lines: I_0, I_1, I_2, I_3
- 2 selection lines: S_0, S_1
- 1 output: Y

Because:

$$2^2 = 4$$

4-to-1 Multiplexer

A 4-to-1 MUX h

- 4 input lines: I_0, I_1, I_2, I_3
- 2 selection lines: S_0, S_1
- 1 output: Y

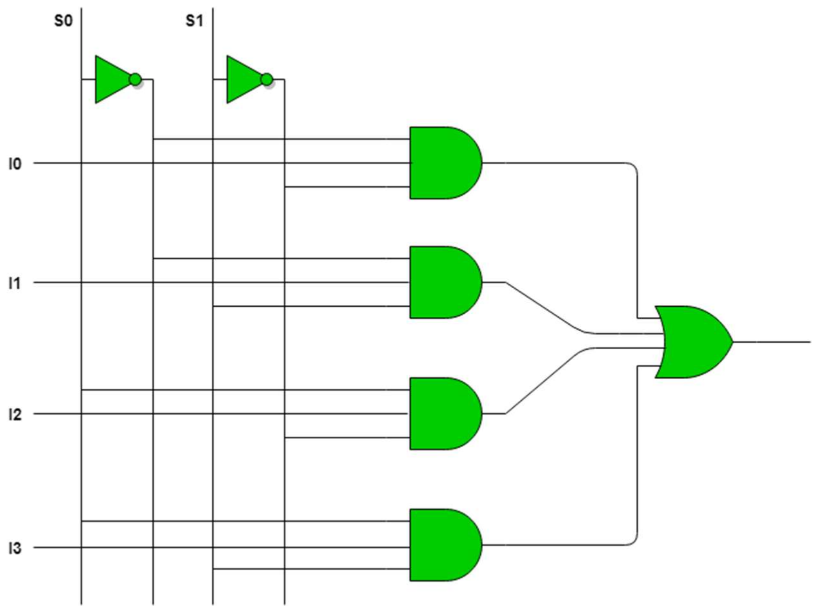
Because:

$$2^2 = 4$$

S_1	S_0	Selected Input	Output Y
0	0	I_0	$Y = I_0$
0	1	I_1	$Y = I_1$
1	0	I_2	$Y = I_2$
1	1	I_3	$Y = I_3$

Boolean Equation

$$Y = \overline{S_1}\,\overline{S_0}I_0 + \overline{S_1}S_0I_1 + S_1\overline{S_0}I_2 + S_1S_0I_3$$



8-to-1 Multiplexer (MUX)

An 8-to-1 MUX has 8 input lines, 3 selection lines, and 1 output line.

$$\boxed{\text{8 Inputs} \rightarrow \text{1 Output}}$$

Because:

$$2^n = 8 \Rightarrow n = 3$$

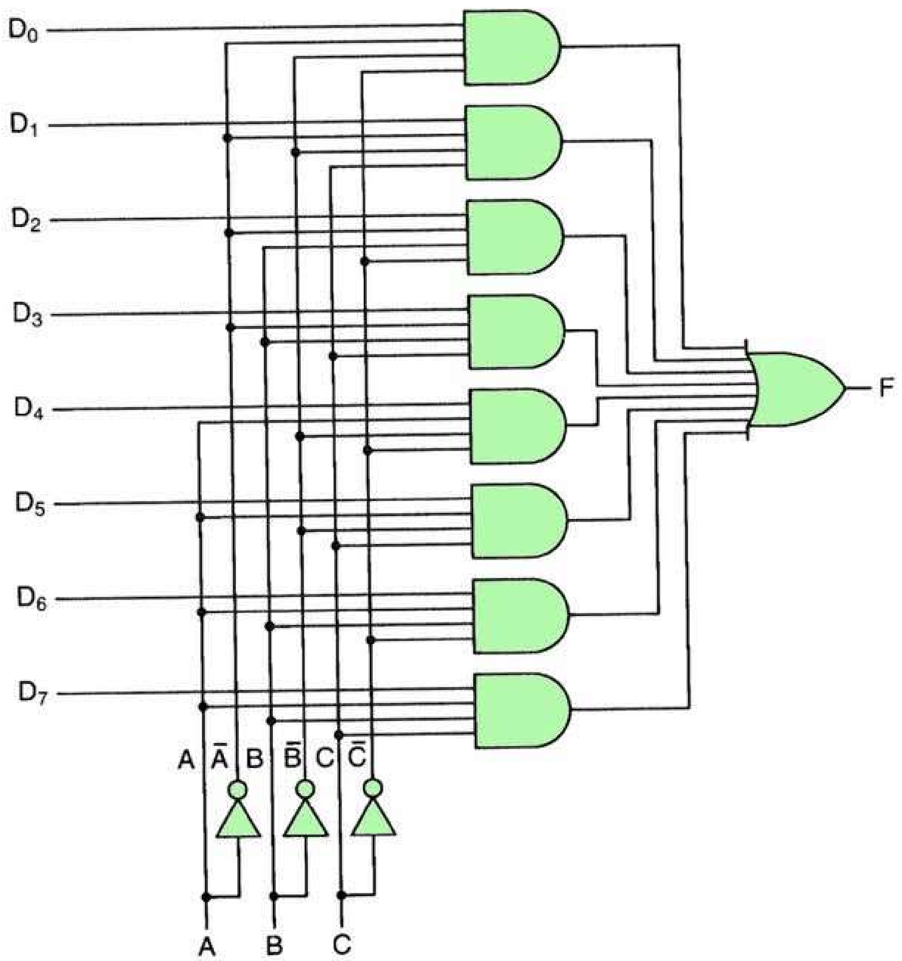
Let:

- Inputs = $I_0, I_1, \dots, I_7$
- Select lines = S_2, S_1, S_0
- Output = Y

S_2	S_1	S_0	Selected Input	Output
0	0	0	I_0	$Y = I_0$
0	0	1	I_1	$Y = I_1$
0	1	0	I_2	$Y = I_2$
0	1	1	I_3	$Y = I_3$
1	0	0	I_4	$Y = I_4$
1	0	1	I_5	$Y = I_5$
1	1	0	I_6	$Y = I_6$
1	1	1	I_7	$Y = I_7$

Boolean Equation

$$\boldsymbol{Y = \overline{S_2 S_1 S_0} I_0 + \overline{S_2 S_1} S_0 I_1 + \overline{S_2} S_1 \overline{S_0} I_2 + \overline{S_2} S_1 S_0 I_3 + S_2 \overline{S_1 S_0} I_4 + S_2 \overline{S_1} S_0 I_5 + S_2 S_1 \overline{S_0} I_6 + S_2 S_1 S_0 I_7}$$



Demultiplexer

A demultiplexer is a combinational logic circuit that takes one input and sends it to one of several output lines according to the select lines.

DEMUX = One input → Many outputs

For a DEMUX with $n$selection lines:

$$\boxed{1 \text{ input} \rightarrow 2^n \text{ outputs}}$$

Example: 1-to-4 DEMUX

- **1 input:** D
- **2 select lines:** S_1, S_0
- **4 outputs:** Y_0, Y_1, Y_2, Y_3

The select lines decide **which output receives the input**.

MUX: Many → One
DEMUX: One → Many

1-to-4 Demultiplexer (DEMUX)

A 1-to-4 DEMUX has:

- 1 input: D
- 2 select lines: S_1, S_0
- 4 outputs**:**

S_1	S_0	**Selected Output**	Y_0	Y_1	Y_2	Y_3
0	0	Y_0	D	0	0	0
0	1	Y_1	0	D	0	0
1	0	Y_2	0	0	D	0
1	1	Y_3	0	0	0	D

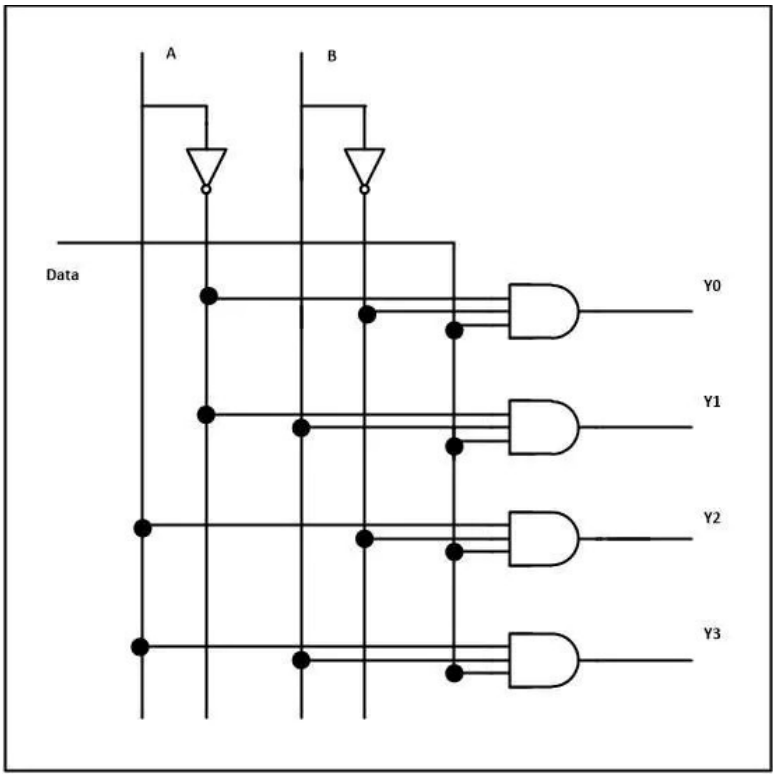
Boolean expression

$$Y_0 = \overline{S_1}\,\overline{S_0}D$$

$$Y_1 = \overline{S_1}S_0D$$

$$Y_2 = S_1\overline{S_0}D$$

$$Y_3 = S_1S_0D$$



1-to-8 DEMUX

A **1-to-8 demultiplexer** has:

- **1 input:** D
- **3 select lines:** S_2, S_1, S_0
- **8 outputs:** $Y_0$to Y_7

S_2	S_1	S_0	**Selected Output**	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
0	0	0	Y_0	D	0	0	0	0	0	0	0
0	0	1	Y_1	0	D	0	0	0	0	0	0
0	1	0	Y_2	0	0	D	0	0	0	0	0
0	1	1	Y_3	0	0	0	D	0	0	0	0
1	0	0	Y_4	0	0	0	0	D	0	0	0
1	0	1	Y_5	0	0	0	0	0	D	0	0
1	1	0	Y_6	0	0	0	0	0	0	D	0
1	1	1	Y_7	0	0	0	0	0	0	0	D

Boolean Equations

$$Y_0 = \overline{S_2}\,\overline{S_1}\,\overline{S_0}D$$

$$Y_1 = \overline{S_2}\,\overline{S_1}S_0D$$

$$Y_2 = \overline{S_2}S_1\overline{S_0}D$$

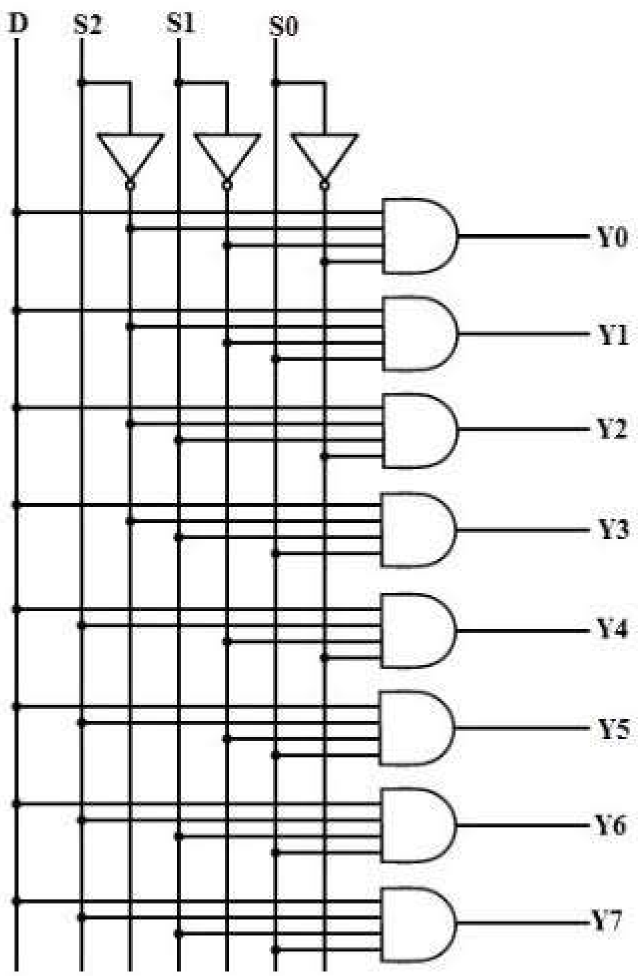
$$Y_3 = \overline{S_2}S_1S_0D$$

$$Y_4 = S_2\overline{S_1}\,\overline{S_0}D$$

$$Y_5 = S_2\overline{S_1}S_0D$$

$$Y_6 = S_2S_1\overline{S_0}D$$

$$Y_7 = S_2S_1S_0D$$



Priority Encoder

A priority encoder is an encoder that produces the binary code of the **highest-priority active input** when multiple inputs are active.

4-to-2 Priority Encoder

It has **4 inputs**: D_3, D_2, D_1, D_0
and **2 outputs**: Y_1, Y_0.

Usually, D_3has the **highest priority** and D_0has the **lowest priority**.

D3	D2	D1	D0	Y1	Y0	Meaning
0	0	0	0	X	X	No input
0	0	0	1	0	0	D0 selected
0	0	1	X	0	1	D1 selected
0	1	X	X	1	0	D2 selected
1	X	X	X	1	1	D3 selected

Simplified Equation:

$$Y_1 = D_3 + D_2$$

$$Y_0 = D_3 + \overline{D_2}D_1$$